

FORM TP 2017207



TEST CODE 02220020

MAY/JUNE 2017

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

ENVIRONMENTAL SCIENCE

AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in THREE sections. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. You may use a silent, non-programmable, scientific calculator to answer questions.
5. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
6. If you use the extra page(s) you **MUST** write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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MODULE 1

Answer BOTH questions.

1. (a) Outline TWO reasons why agriculture is important in the Caribbean.

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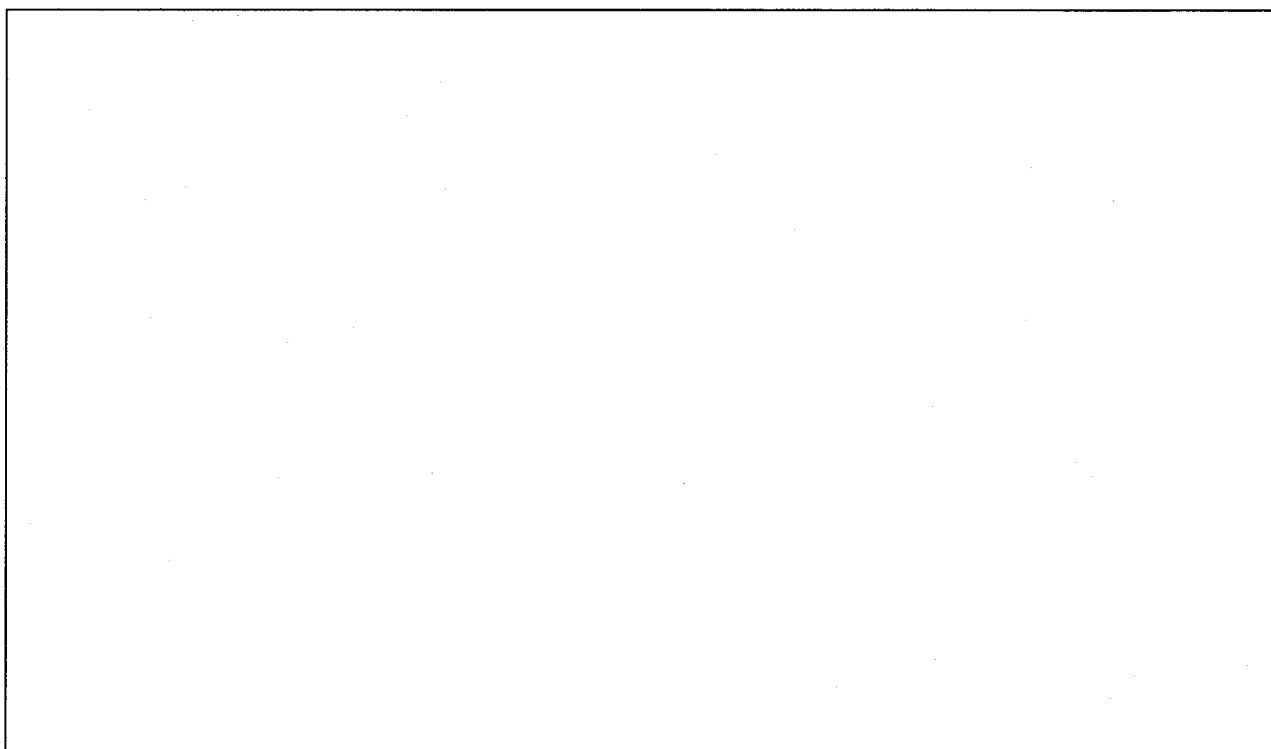
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[4 marks]

- (b) (i) Draw a diagram to illustrate the environmentally sustainable practice of terracing in agricultural systems.



[3 marks]

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- (ii) Name ONE other environmentally sustainable practice used in agricultural systems.

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[1 mark]

- (c) Outline THREE ways in which genetic engineering can be used to improve agriculture.

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[6 marks]

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- (d) Discuss TWO negative effects of using agrochemicals.

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[6 marks]

Total 20 marks



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2. An environmental science student proposed the device in Figure 1 as a model for a sustainable agriculture project. The system is manually irrigated daily with a fertilizer solution and flushed weekly with tap water.

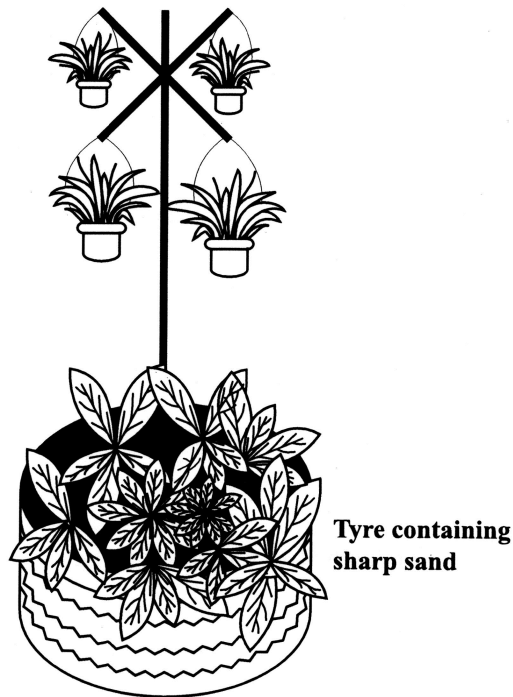


Figure 1. Sustainable agriculture method



- (a) Discuss THREE advantages of using the device shown in Figure 1.

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[6 marks]

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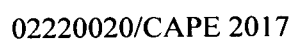
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- [3 marks]**

- [3 marks]**

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MODULE 2

Answer **BOTH** questions.

3. Figure 2 is a pie chart showing the energy usage in a Caribbean country. The total energy usage for the country is 1.84×10^7 gigajoules (GJ).

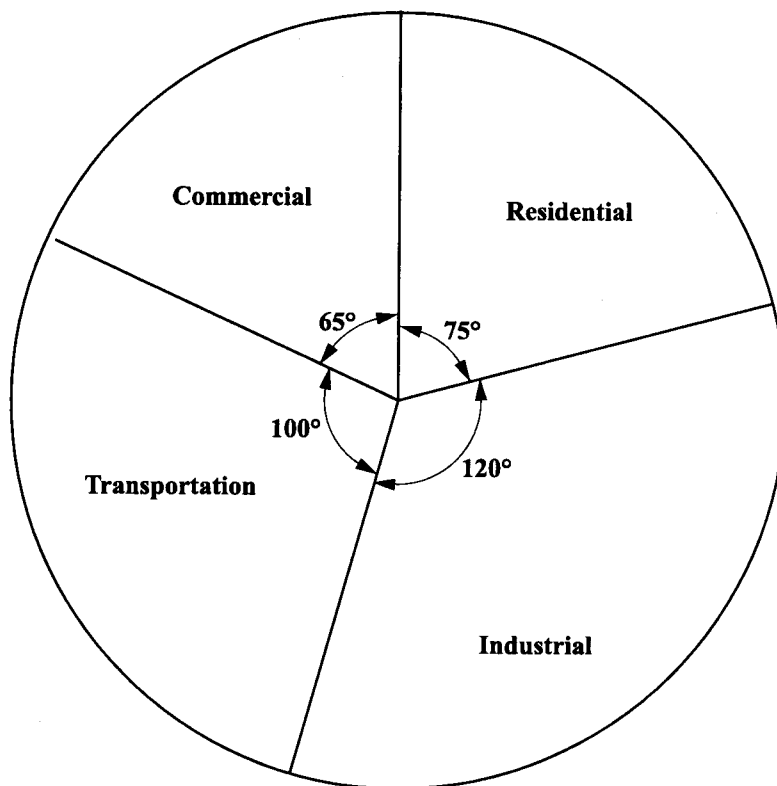


Figure 2. Energy use in a Caribbean country

- (a) (i) Calculate, showing all working, the percentage of energy in this country that is used for the transportation sector.

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[2 marks]



- (ii) Given that the total energy consumed in this Caribbean country is 1.84×10^7 gigajoules, calculate, showing all working, the total amount of energy consumed by the residential and industrial sectors.

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[4 marks]

- (b) (i) Distinguish between 'energy conservation' and 'energy efficiency'.

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[2 marks]

- (ii) State TWO energy conservation methods that may be used in the transportation sector.

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[2 marks]

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- [illegible]

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[illegible]

Total 20 marks



RESEARCH DESIGN AND METHODS

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4. (a) Electricity is described as a secondary energy source. State ONE primary energy source and distinguish between 'primary' and 'secondary' sources of energy.

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[3 marks]

- (b) At a conference, a speaker states that "the world's socioeconomic dependency on fossil fuel combustion will make it difficult to reduce carbon dioxide emissions". Explain this statement.

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[3 marks]

- (c) (i) Define the term 'efficiency of conversion' as it refers to energy.

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[2 marks]



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- (d) The graph below shows the cost of gasoline in a Caribbean country from 1975 to 2015.

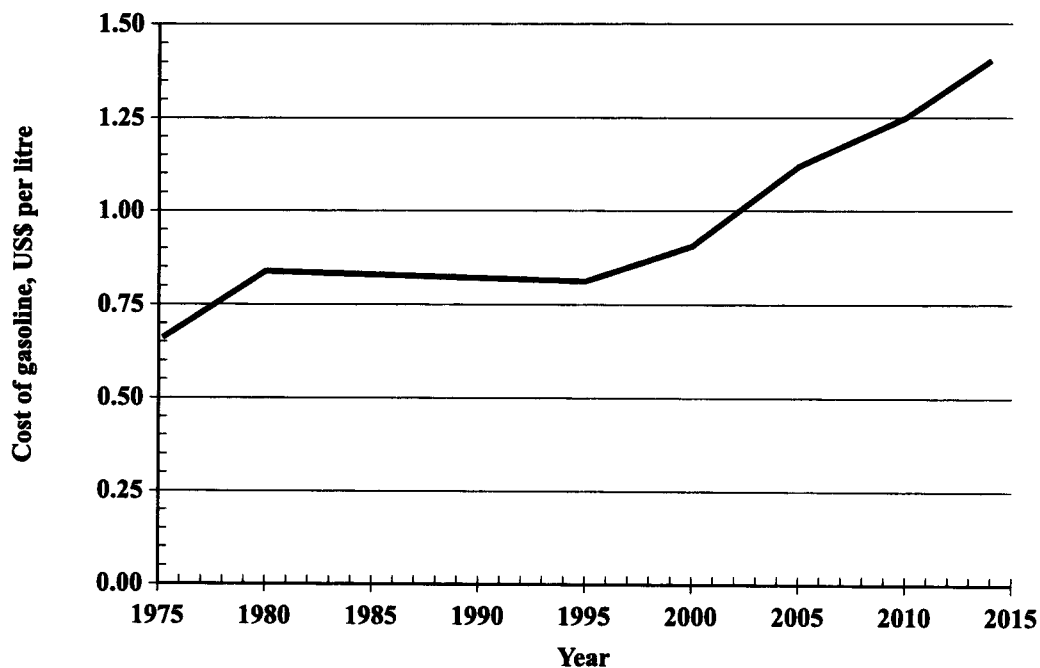


Figure 3. Cost of gasoline in a Caribbean country from 1975 to 2015

- (i) What was the price of gasoline in 2000?

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[1 mark]

- (ii) Describe the OVERALL trend illustrated in the graph.

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[3 marks]



- [illegible]

Total 20 marks



1. The first step in the process is to identify the problem or issue that needs to be addressed. This involves gathering information and understanding the context of the problem.

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MODULE 3

Answer BOTH questions.

5. Table 1 shows the amount of different types of municipal waste generated by a country.

TABLE 1: MUNICIPAL WASTE GENERATED BY A COUNTRY

Type of Waste	Tonnes
Paper	200
Glass	50
Plastic	500
Aluminium	300

- (a) Draw a bar graph on the grid provided on page 21 to illustrate the information in Table 1. [5 marks]
- (b) Use suitable equations to explain how the use of chlorofluorocarbons (CFCs) destroys the ozone layer.

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- (c) State THREE health risks associated with ozone depletion.

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[3 marks]

- (d) Explain how the Montreal Protocol helps to reduce ozone depletion.

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[6 marks]



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6. Figure 4 is a diagram showing a cross section of a solid waste disposal site and Figure 5 is a diagram showing an aerial view of the same solid waste disposal site.

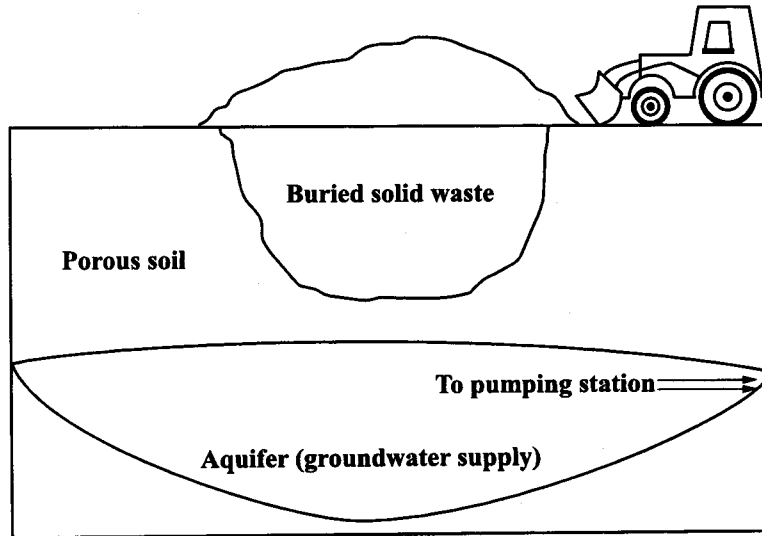


Figure 4. Cross section of solid waste disposal site

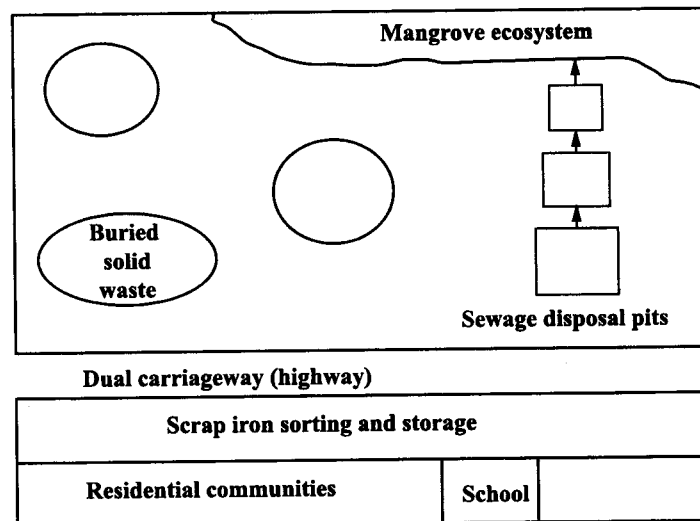


Figure 5. Aerial view of solid waste disposal site

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- [illegible]

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[6 marks]

- (b) (i) Identify TWO critical parts that are missing from the landfill in Figure 4.

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[2 marks]

- (ii) Recycling is one mechanism that can reduce land pollution from landfills by reducing waste materials in the landfill. Outline THREE critical steps that are required for a recycling project to work.

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[3 marks]

- (iii) Evaluate whether large-scale recycling would be feasible in the Caribbean context.

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[3 marks]

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- (c) An environmental conservation group recommended incineration as an alternative method for treating the hazardous waste which is brought to the site in Figure 4. State THREE advantages and THREE disadvantages of incineration.

Advantages

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Disadvantages

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[6 marks]

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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